



SQL Data Types Practice Assessment

Instructions

- Complete all exercises in MySQL Workbench
- Copy and paste your SQL commands into a document
- Take screenshots of your results

Part 1: Simple Table Creation

Exercise 1: Student Information Table

Create a table called students to store student information.

What you need to store:

- Student ID (whole numbers, automatically increases)
- Student name (different length names like "Ali" or "Muhammad Abdullah")
- Roll number (always 6 characters like "CS0001")
- Email address (different lengths)
- Age (whole numbers only)
- CGPA (numbers like 3.75, exact values needed)
- Is scholarship student? (Yes/No)
- Date of birth (dates only)
- Phone number (always 11 digits like "03001234567")

Your Task: Write the CREATE TABLE command with appropriate data types.



Exercise 2: Library Books Table

Create a table called books for a library system.

What you need to store:

- Book ID (whole numbers, starts from 1)
- Book title (variable length)
- Author name (variable length up to 80 characters)
- ISBN (always 13 characters like "9781234567890")
- Category (must be one of: Fiction, Non-Fiction, Science, History)
- Price (exact money values like 599.99)
- Number of pages (whole numbers)
- Is available for lending? (Yes/No)
- Publication date (dates only)
- Brief description (long text)

Your Task: Write the CREATE TABLE command.

Part 2: Understanding Different Number Types

Exercise 3: Online Store Orders

Create a table called orders that handles different types of numbers.

What you need to store:

- Order ID (very large numbers for millions of orders)
- Customer ID (regular whole numbers)
- Order amount (exact money - no rounding errors!)
- Tax amount (exact money)
- Shipping weight (approximate decimals like 2.567 kg)
- Number of items (small whole numbers 1-50)
- Is express delivery? (Yes/No)
- Order date and time together
- Discount percentage (approximate decimals)



Your Task:

1. Create the table with proper numeric data types
2. Insert these 3 sample orders:

Order 1: Customer 101, Amount 1250.75, Tax 125.08, Weight 3.2kg, 5 items, Express delivery, Today's date

Order 2: Customer 205, Amount 89.99, Tax 9.00, Weight 0.5kg, 1 item, Regular delivery, Yesterday

Order 3: Customer 156, Amount 2999.50, Tax 299.95, Weight 8.7kg, 12 items, Express delivery, Today

Questions to Answer:

1. Why did you choose DECIMAL for money instead of FLOAT?
2. Why use BIGINT for Order ID?
3. What happens if you try to insert a negative number of items?

Part 3: Text and String Types

Exercise 4: User Registration System

Create a users table that shows the difference between CHAR and VARCHAR.

What you need to store:

- User ID (whole numbers)
- Username (3 to 25 characters)
- Country code (always 2 letters like "PK", "US")
- State code (always 3 letters like "SND", "PUN")
- Full name (variable length up to 100 characters)
- Password (always 32 characters - encrypted)
- Account type (Premium, Basic, Guest)
- About me (long biographical text)
- Verification code (always 6 digits like "123456")

Your Task:

1. Create the table using CHAR for fixed-length fields and VARCHAR for variable-length fields
2. Insert 5 different users with varying name lengths



Test Understanding: Insert a user with username "Ali" into your table, then insert another with username "MuhammadAbdullah". Explain the storage difference.

Part 4: Dates and Times

Exercise 5: Event Management

Create an events table for managing school events.

What you need to store:

- Event ID (whole numbers)
- Event name (variable text)
- Event date (date only)
- Start time (time only)
- End time (time only)
- Registration deadline (date and time together)
- Created timestamp (automatically set when record is added)
- Is public event? (Yes/No)
- Max participants (small whole numbers)

Your Task:

1. Create the table
2. Insert 3 upcoming events with different dates and times
3. Write a query to find events happening today

Think About These Questions

1. When would you use CHAR instead of VARCHAR?
2. Why is DECIMAL better than FLOAT for storing prices?
3. What's the difference between DATE and DATETIME?
4. When should you use ENUM data type?
5. What happens if you try to store text in an INT column?



Helpful Tips

- **For money:** Always use DECIMAL(10,2)
- **For IDs:** Use INT with AUTO_INCREMENT
- **For Yes/No:** Use BOOLEAN or TINYINT(1)
- **For fixed codes:** Use CHAR(n)
- **For variable names:** Use VARCHAR(n)
- **For long text:** Use TEXT
- **For timestamps:** Use TIMESTAMP with DEFAULT CURRENT_TIMESTAMP

Good Luck!

Remember: The goal is to understand which data type fits which kind of information. Think about the data you're storing and choose accordingly!